

L06 – Graph Neural Networks (GNN)

Foundations Lecture Series

Course Instructor

Microgrid 3-phase Security AI Project

Foundations Lecture 6

Keywords: message passing, GCN, GraphSAGE, GAT, node/edge/graph-level tasks

Lecture roadmap

- 1 Why graphs and why an MLP is not enough
- 2 Message passing: the core idea
- 3 Major variants: GCN, GraphSAGE, GAT
- 4 Tasks: node-level, edge-level, graph-level
- 5 Known limitations
- 6 Power-grid application preview

Prerequisite

L05 (MLPs and PyTorch). Helpful: basic graph theory (nodes, edges, adjacency matrix).

Examples of graph-structured data

TODO: *Bullet list: molecules, citation networks, social graphs, transportation networks, power grids.
Visual of a power grid as a graph with buses=nodes, lines=edges.*

Why an MLP on flattened features is not enough

TODO: *Show how flattening a graph into a fixed-size feature vector loses topology. An MLP cannot leverage the fact that bus 3 is two hops from bus 1. Set up the need for message passing.*

The core idea in three steps

TODO: (1) Each node has a hidden state. (2) Each node receives messages from its neighbours. (3) Each node updates its state by aggregating messages and combining with its own. Repeat for L layers. Diagram: 2-hop neighborhood.

The general message-passing equation

TODO: Write $h_v^{(l+1)} = \text{UPDATE}(h_v^{(l)}, \text{AGG}(\{h_u^{(l)} : u \in \mathcal{N}(v)\}))$. Explain each piece. Note that different choices of AGG and UPDATE give different GNN variants.

GCN (Kipf & Welling, 2017)

TODO: Write the GCN equation: $H^{(l+1)} = \sigma(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(l)} W^{(l)})$. Symmetric normalization.
Note: simple, effective baseline. PyTorch Geometric implementation is one line.

GraphSAGE (Hamilton et al., 2017)

TODO: *Mean / max / LSTM aggregator. Inductive (handles unseen nodes), unlike vanilla GCN which is transductive. Practical choice when test graphs differ from train.*

TODO: *Attention-weighted message passing. Each neighbour gets a learned attention coefficient α_{ij} . Worth the extra parameters when some neighbours matter much more than others. Multi-head attention typically.*

Node-level, edge-level, graph-level outputs

TODO: *Three task families with one example each. For our project, we mostly want graph-level risk scores: “does this microgrid state + contingency lead to a violation?”. Note that pooling (mean/sum/max) collapses node embeddings to a single graph vector.*

The project's GNN

TODO: *Forward-reference: Week 7 builds a phase-aware GNN with bus nodes, line edges, contingency masks, and graph-level readout. Honest note: on 77 samples it does not beat a flat MLP. We will discuss why.*

Oversmoothing

TODO: *After many layers, all node embeddings converge to the same vector. Limits effective depth to 2–4 layers. Mitigations: residual connections, GraphSAGE max-pool, recent “deep GNN” research.*

Expressivity bounds

TODO: *Brief note on the Weisfeiler-Lehman test and what GNNs can/cannot distinguish. Show two non-isomorphic graphs that a standard GNN cannot tell apart. Caveat: not load-bearing for our application, but good to know.*

Small-data behavior

TODO: *GNNs typically need more data than MLPs to show their value, because the message-passing weights are shared across graphs. On 77 samples, MLPs often win. Connect to the project's honest negative result.*

What a power-grid GNN looks like

TODO: Walk through: nodes = buses, edges = lines/transformers, node features = per-phase load/PV/voltage, edge features = R/X/rating, global features = scenario metadata, output = graph-level risk score. This is the Week 7 architecture.

TODO: *Suggest: implement a 2-layer GCN on Cora citation network with PyTorch Geometric, achieve $>80\%$ test accuracy. Then visualize node embeddings with t-SNE before and after training.*

TODO: *Items: (1) write the message-passing equation, (2) name three GNN variants and their key differences, (3) explain oversmoothing in one sentence, (4) match node/edge/graph-level tasks to power-grid examples.*

TODO: *Hamilton's Graph Representation Learning book (free online), Kipf & Welling 2017 (GCN), Hamilton et al. 2017 (GraphSAGE), Veličković et al. 2018 (GAT), Xu et al. 2019 (Weisfeiler-Lehman bounds).*